

Adaptive double-phase ROF model: report on cross-dataset evaluation on grayscale images

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Abstract

This note reports the results of numerical experiments concerning the cross-dataset performance of the adaptive double-phase ROF model for grayscale images, using parameters obtained from BSDS500 without dataset-specific retuning.

1 Cross-dataset performance

To evaluate the generalization of the proposed parameter selection, we test the parameters obtained from BSDS500 on four unseen datasets: Kodak24, Urban100, EuroSAT and Set14. Each one of these datasets consists of color images. Here we work with their grayscale versions. The case of actual color images is discussed in the 'RGB' supplement.

The parameters are transferred directly from BSDS500 without any dataset-specific retuning. The experiments are performed for all considered noise levels, and the resulting SSIM, PSNR, and LPIPS values are reported below. In particular, the transferred prescription is given by $\lambda = 0.5\sigma$ and $a = 60$, while b/a is set to 100, 50, or 25, depending on the considered noise regime. In addition, we examine the sensitivity of the model to perturbations of the transferred parameters. We observe that, on average, the YUV-dpROF and RGB-dpROF models were performing significantly worse already than the vectorial dpROF already on the BSDS500 dataset. From now on we only consider the vectorial dpROF model and we test its performance over several unseen datasets below.

Regarding the datasets used for evaluation, Kodak24 is a standard benchmark dataset for image restoration and denoising (available at [2]). It consists of 24 high-quality natural color images of size 768×512 pixels, containing a variety of natural scenes, textures, smooth regions, and sharp edges. Compared with BSDS500, it provides higher-resolution images and a different collection of natural image content.

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On the other hand, Urban100 [6] (available here [4]) is a benchmark consisting of urban scenes. Its images contain repetitive patterns, fine structures, sharp edges, and highly textured regions. Hence the provided image characteristics are substantially different from those typically encountered in BSDS500.

The third dataset considered is EuroSAT [5] (available at [1]). This dataset consists of 27000 very small (64×64 pixels) images, which are patches of terrain of different types (forests, rivers, industrial etc.) with very different features. Here, both the resolution of the image and the image characteristics are very different to those in the training dataset; note that here we do not adapt the model to deal with specific noise types for satellite imaging, but again only use the 'black-box' version of our model.

Finally, Set14 is a classical dataset for image processing containing mostly natural images. We sourced it from [3], where it is available along with a 4x upscaled version. We use it for to check the behavior of our model in higher resolution, or more precisely: on different resolutions of the same set of images.

Table 1: Cross-dataset performance of dpROF using the parameter prescription derived from BSDS500. No dataset-specific parameter retuning was performed on Kodak24, Urban100, EuroSAT or Set14 datasets.

Dataset	Metric	Noise variance σ^2							
		0.001	0.002	0.004	0.007	0.01	0.02	0.04	0.07
BSDS500	SSIM $\uparrow$	0.9206	0.8871	0.8411	0.8014	0.7720	0.7025	0.6412	0.5892
	LPIPS $\downarrow$	0.0823	0.1240	0.1815	0.2263	0.2618	0.3274	0.3910	0.4441
	PSNR $\uparrow$	33.32	31.45	29.61	28.23	27.38	25.74	24.20	22.77
Kodak24	SSIM $\uparrow$	0.9118	0.8779	0.8324	0.7953	0.7677	0.7063	0.6526	0.6096
	LPIPS $\downarrow$	0.0966	0.1435	0.2100	0.2554	0.2941	0.3611	0.4278	0.4799
	PSNR $\uparrow$	34.03	32.19	30.37	29.04	28.19	26.56	25.04	23.63
Urban100	SSIM $\uparrow$	0.9362	0.9067	0.8660	0.8231	0.7946	0.7118	0.6393	0.5724
	LPIPS $\downarrow$	0.0427	0.0727	0.1208	0.1676	0.2044	0.2885	0.3730	0.4446
	PSNR $\uparrow$	32.58	30.47	28.42	26.70	25.76	23.72	22.05	20.61
EuroSAT	SSIM $\uparrow$	0.9189	0.8860	0.8388	0.8096	0.7816	0.7325	0.6863	0.6492
	LPIPS $\downarrow$	0.0610	0.0923	0.1413	0.1629	0.1879	0.2362	0.2695	0.2977
	PSNR $\uparrow$	36.05	34.31	32.43	31.60	30.73	29.77	28.28	26.62
Set14 (low)	SSIM $\uparrow$	0.9154	0.8825	0.8375	0.7984	0.7700	0.7045	0.6432	0.5917
	LPIPS $\downarrow$	0.0641	0.0985	0.1503	0.2008	0.2363	0.3026	0.3767	0.4310
	PSNR $\uparrow$	33.39	31.52	29.62	28.23	27.33	25.63	23.91	22.37
Set14 (high)	SSIM $\uparrow$	0.9375	0.9101	0.8697	0.8570	0.8333	0.8057	0.7643	0.7321
	LPIPS $\downarrow$	0.1377	0.2050	0.2933	0.2991	0.3422	0.3478	0.4022	0.4403
	PSNR $\uparrow$	37.16	35.15	33.00	31.80	30.69	28.91	26.60	24.48

The results show that the derived BSDS500 parameter selection can be directly applied to the unseen datasets over the whole considered noise range and without any dataset specific re-tuning. In particular, no parameters are selected or adjusted when testing on Kodak24, Urban100, or Set14. Thus, these experiments provide a direct evaluation of the cross-dataset transfer of the proposed parameter selection. The results indicate that the BSDS500-derived

parameter prescription remains effective on datasets with different image characteristics. Although the absolute metric values vary across datasets, the overall behavior of the model is preserved over the considered noise levels. This suggests that the selected parameters are not strongly tied to the specific image statistics of BSDS500 and show good cross-dataset generalization within the considered natural-image benchmarks.

To further assess the cross-dataset behavior of the model, we compare dpROF with the classical ROF model, using in both cases the corresponding parameter prescriptions derived from BSDS500 and without any dataset-specific retuning. On Kodak24, dpROF achieves higher SSIM and PSNR than ROF at 8/8 considered noise levels, while lower LPIPS values are obtained at 6/8 noise levels. On Urban100, the improvement in SSIM and PSNR is again preserved across the entire noise range, whereas the LPIPS comparison is less uniform. On the EuroSAT dataset, dpROF achieves higher SSIM and PSNR than ROF at 5/8 considered noise levels, while lower LPIPS values are obtained at 6/8 noise levels. Overall, across Kodak24, Urban100 and EuroSAT, dpROF outperforms ROF in terms of SSIM and PSNR in all 24/24 considered dataset and noise-level combinations. These results show that the improvements observed on BSDS500 are preserved when the parameter prescriptions are transferred to unseen natural-image datasets, especially with respect to SSIM and PSNR.

Finally, we verify how the dpROF model performs on the same pictures across different image resolutions. To this end, we used the Set14 dataset in two sizes: the originals and the 4x upscaled versions available here [3]. We see that the parameter prescriptions derived from BSDS500, without any dataset-specific retuning, perform well for both low and high resolutions: for the low resolution images, dpROF achieves higher SSIM and PSNR than ROF at 8/8 considered noise levels, while lower LPIPS values are obtained at 2/8 noise levels. For the high resolution images, dpROF outperforms ROF in all metrics at 5/8 noise levels: at the lowest noise levels, ROF slightly outperforms dpROF, but starting with the noise level 0.007 dpROF consistently outperforms ROF by a significant margin. This suggests that possibly the method can be further improved at low noise levels by modifying the parameters for high image resolutions, but even using the BSDS500-derived parameters the method is robust.

Table 2: Comparison of dpROF and ROF on Kodak24 using the corresponding BSDS500-derived parameter prescriptions without dataset-specific retuning.

Method	Metric	Noise variance σ^2							
		0.001	0.002	0.004	0.007	0.01	0.02	0.04	0.07
dpROF	SSIM $\uparrow$	0.9118	0.8779	0.8324	0.7953	0.7677	0.7063	0.6526	0.6096
	LPIPS $\downarrow$	0.0966	0.1435	0.2100	0.2554	0.2941	0.3611	0.4278	0.4799
	PSNR $\uparrow$	34.03	32.19	30.37	29.04	28.19	26.56	25.04	23.63
ROF	SSIM $\uparrow$	0.9074	0.8721	0.8273	0.7832	0.7512	0.7011	0.6411	0.5985
	LPIPS $\downarrow$	0.0978	0.1424	0.2039	0.2635	0.3043	0.3674	0.4409	0.4924
	PSNR $\uparrow$	33.72	31.94	30.24	28.89	28.05	26.47	24.88	23.46

Table 3: Comparison of dpROF and ROF on Urban100 using the corresponding BSDS500-derived parameter prescriptions without dataset-specific retuning.

Method	Metric	Noise variance σ^2							
		0.001	0.002	0.004	0.007	0.01	0.02	0.04	0.07
dpROF	SSIM $\uparrow$	0.9362	0.9067	0.8660	0.8231	0.7946	0.7118	0.6393	0.5724
	LPIPS $\downarrow$	0.0427	0.0727	0.1208	0.1676	0.2044	0.2885	0.3730	0.4446
	PSNR $\uparrow$	32.58	30.47	28.42	26.70	25.76	23.72	22.05	20.61
ROF	SSIM $\uparrow$	0.9334	0.9018	0.8594	0.8160	0.7838	0.7097	0.6290	0.5582
	LPIPS $\downarrow$	0.0431	0.0731	0.1187	0.1685	0.2063	0.2815	0.3721	0.4458
	PSNR $\uparrow$	32.22	30.11	28.10	26.56	25.61	23.59	21.82	20.33

Table 4: Comparison of dpROF and ROF on EuroSAT using the corresponding BSDS500-derived parameter prescriptions without dataset-specific retuning.

Method	Metric	Noise variance σ^2							
		0.001	0.002	0.004	0.007	0.01	0.02	0.04	0.07
dpROF	SSIM $\uparrow$	0.9189	0.8860	0.8388	0.8096	0.7816	0.7325	0.6863	0.6492
	LPIPS $\downarrow$	0.0610	0.0923	0.1413	0.1629	0.1879	0.2362	0.2695	0.2977
	PSNR $\uparrow$	36.05	34.31	32.43	31.60	30.73	29.77	28.28	26.62
ROF	SSIM $\uparrow$	0.9154	0.8815	0.8365	0.7913	0.7592	0.7274	0.6757	0.6410
	LPIPS $\downarrow$	0.0620	0.0940	0.1393	0.1832	0.2132	0.2280	0.2690	0.2941
	PSNR $\uparrow$	35.96	34.28	32.56	31.19	30.33	29.59	28.02	26.45

Table 5: Comparison of ROF and dpROF on Set14 (low resolution) using the corresponding BSDS500-derived parameter prescriptions without dataset-specific retuning.

Method	Metric	Noise variance σ^2							
		0.001	0.002	0.004	0.007	0.01	0.02	0.04	0.07
dpROF	SSIM $\uparrow$	0.9154	0.8825	0.8375	0.7984	0.7700	0.7045	0.6432	0.5917
	LPIPS $\downarrow$	0.0641	0.0985	0.1503	0.2008	0.2363	0.3026	0.3767	0.4310
	PSNR $\uparrow$	33.39	31.52	29.62	28.23	27.33	25.63	23.91	22.37
ROF	SSIM $\uparrow$	0.9112	0.8776	0.8337	0.7906	0.7600	0.6996	0.6333	0.5814
	LPIPS $\downarrow$	0.0645	0.0960	0.1412	0.1903	0.2232	0.2953	0.3740	0.4318
	PSNR $\uparrow$	33.15	31.34	29.56	28.15	27.26	25.50	23.73	22.16

Table 6: Comparison of ROF and dpROF on Set14 (high resolution) using the corresponding BSDS500-derived parameter prescriptions without dataset-specific retuning.

Method	Metric	Noise variance σ^2							
		0.001	0.002	0.004	0.007	0.01	0.02	0.04	0.07
dpROF	SSIM $\uparrow$	0.9375	0.9101	0.8697	0.8570	0.8333	0.8057	0.7643	0.7321
	LPIPS $\downarrow$	0.1377	0.2050	0.2933	0.2991	0.3422	0.3478	0.4022	0.4403
	PSNR $\uparrow$	37.16	35.15	33.00	31.80	30.69	28.91	26.60	24.48
ROF	SSIM $\uparrow$	0.9386	0.9120	0.8753	0.8370	0.8086	0.7952	0.7485	0.7200
	LPIPS $\downarrow$	0.1292	0.1903	0.2659	0.3280	0.3668	0.3856	0.4439	0.4753
	PSNR $\uparrow$	37.24	35.30	33.27	31.57	30.48	28.71	26.37	24.26

1.1 Parameter sensitivity on Urban100

To further examine the robustness of the transferred parameter prescription, we perform a local one-at-a-time sensitivity analysis on Urban100. Each parameter is varied around its transferred value while the remaining parameters are kept fixed. This allows us to assess how moderate parameter perturbations affect the reconstruction performance without introducing any dataset-specific retuning.

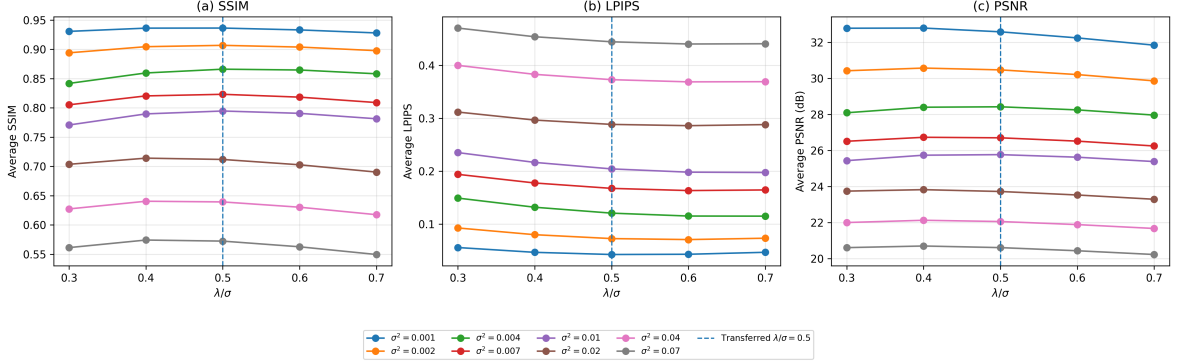


Figure 1: Sensitivity of dpROF with respect to the regularization parameter λ on Urban100. The vertical dashed line indicates the transferred BSDS500-derived prescription $\lambda/\sigma = 0.5$. For each noise level, λ/σ is varied while the remaining parameters are kept fixed at their corresponding transferred values.

Table 7: Sensitivity of dpROF with respect to the regularization parameter λ on Urban100. The remaining parameters are kept fixed at their transferred values.

Noise variance σ^2	λ/σ	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$
0.001	0.3	0.930567	0.055845	32.7756
	0.4	0.936090	0.046777	32.7849
	0.5 [†]	0.936229	0.042673	32.5767
	0.6	0.933062	0.043128	32.2395
	0.7	0.927904	0.046851	31.8366
0.002	0.3	0.893961	0.092842	30.4167
	0.4	0.904433	0.080099	30.5657
	0.5 [†]	0.906726	0.072747	30.4651
	0.6	0.903791	0.070889	30.2048
	0.7	0.897632	0.073438	29.8556
0.004	0.3	0.841717	0.149274	28.0867
	0.4	0.859551	0.131992	28.3935
	0.5 [†]	0.866005	0.120824	28.4173
	0.6	0.864620	0.115354	28.2458
	0.7	0.858134	0.115152	27.9571
0.007	0.3	0.805200	0.194119	26.5054
	0.4	0.820374	0.177808	26.7262
	0.5 [†]	0.823107	0.167621	26.6968
	0.6	0.818245	0.163479	26.5126
	0.7	0.808994	0.164661	26.2427
0.010	0.3	0.770711	0.235232	25.4310
	0.4	0.789650	0.216693	25.7323
	0.5 [†]	0.794569	0.204406	25.7638
	0.6	0.790595	0.198255	25.6216
	0.7	0.781309	0.197614	25.3797
0.020	0.3	0.703564	0.311834	23.7394
	0.4	0.714062	0.296675	23.8207
	0.5 [†]	0.711839	0.288516	23.7193
	0.6	0.702710	0.286105	23.5251
	0.7	0.690257	0.288159	23.2881
0.040	0.3	0.627353	0.400020	21.9972
	0.4	0.640463	0.383057	22.1225
	0.5 [†]	0.639308	0.373033	22.0524
	0.6	0.630283	0.368885	21.8830
	0.7	0.617488	0.369224	21.6679
0.070	0.3	0.561278	0.470526	20.6049
	0.4	0.574212	0.454120	20.6953
	0.5 [†]	0.572380	0.444579	20.6057
	0.6	0.562675	0.440404	20.4320
	0.7	0.549502	0.440895	20.2237

[†] Transferred BSDS500-derived reference value.

Table 8: Sensitivity of dpROF with respect to the parameter a on Urban100.

σ^2	a	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$
0.001	40	0.936589	0.043507	32.6600
	60 [†]	0.936229	0.042673	32.5767
	80	0.935632	0.042445	32.5015
0.002	40	0.906312	0.074613	30.5110
	60 [†]	0.906726	0.072747	30.4651
	80	0.906617	0.071766	30.4223
0.004	40	0.864403	0.123196	28.4342
	60 [†]	0.866005	0.120824	28.4173
	80	0.866676	0.119363	28.3986
0.007	40	0.822606	0.170150	26.7489
	60 [†]	0.823107	0.167621	26.6968
	80	0.822525	0.166436	26.6500
0.010	40	0.793044	0.207492	25.7947
	60 [†]	0.794569	0.204406	25.7638
	80	0.794659	0.202770	25.7332
0.020	40	0.714920	0.288928	23.8136
	60 [†]	0.711839	0.288516	23.7193
	80	0.708028	0.289663	23.6399
0.040	40	0.641149	0.374593	22.1168
	60 [†]	0.639308	0.373033	22.0524
	80	0.636325	0.373428	21.9952
0.070	40	0.574168	0.446260	20.6639
	60 [†]	0.572380	0.444579	20.6057
	80	0.569519	0.444827	20.5543

[†] Transferred BSDS500-derived reference value.

Table 9: Sensitivity of dpROF with respect to the ratio b/a on Urban100.

σ^2	b/a	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$
0.001	60	0.931306	0.047026	32.1988
	100 [†]	0.936229	0.042673	32.5767
	140	0.937537	0.042205	32.7068
0.002	60	0.902078	0.075620	30.2020
	100 [†]	0.906726	0.072747	30.4651
	140	0.906958	0.074490	30.5399
0.004	60	0.863935	0.118934	28.2737
	100 [†]	0.866005	0.120824	28.4173
	140	0.863540	0.125181	28.4361
0.007	30	0.810959	0.174632	26.3926
	50 [†]	0.823107	0.167621	26.6968
	70	0.825143	0.169373	26.7825
0.010	30	0.783963	0.207446	25.5402
	50 [†]	0.794569	0.204406	25.7638
	70	0.794264	0.208716	25.8033
0.020	15	0.686660	0.305061	23.2754
	25 [†]	0.711839	0.288516	23.7193
	35	0.722927	0.284439	23.8800
0.040	15	0.614317	0.386185	21.7094
	25 [†]	0.639308	0.373033	22.0524
	35	0.648979	0.372685	22.1500
0.070	15	0.547877	0.458020	20.2954
	25 [†]	0.572380	0.444579	20.6057
	35	0.581712	0.445256	20.6900

[†] Transferred BSDS500-derived reference value.

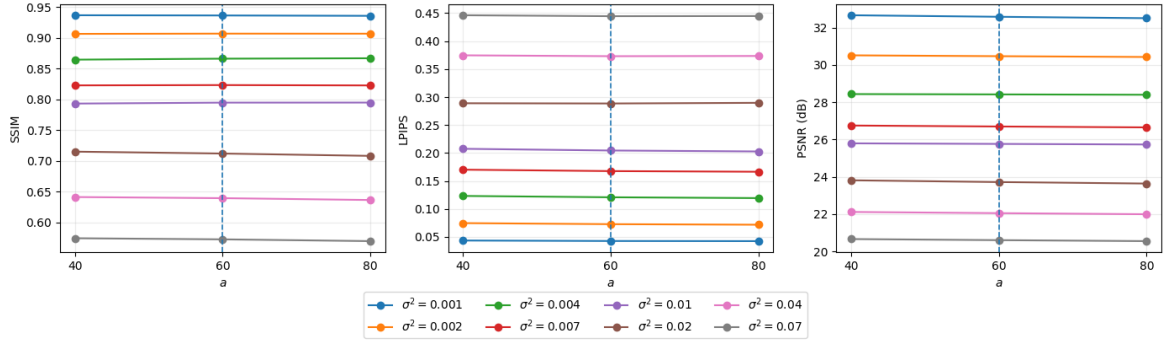


Figure 2: Sensitivity of dpROF with respect to the parameter a on Urban100. The vertical dashed line indicates the transferred value $a = 60$.

The sensitivity analysis shows different behaviors for the considered parameters. The dependence on a is relatively weak in the examined range, since moderate perturbations around the transferred value $a = 60$ lead only to small changes in SSIM, LPIPS, and PSNR. The ratio b/a also show small changes at several low and medium noise levels, while a more noticeable and asymmetric influence appears at the higher noise levels. In particular, decreasing the

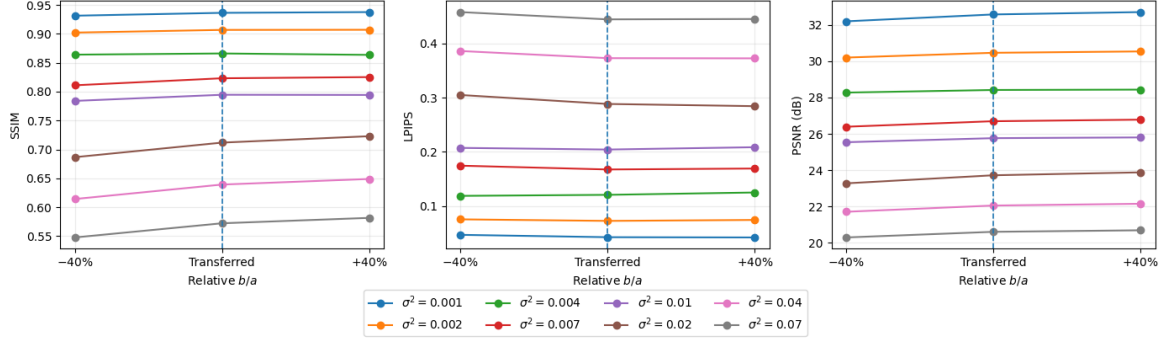


Figure 3: Sensitivity of dpROF with respect to the ratio b/a on Urban100. The vertical dashed line indicates the transferred value. The horizontal axis shows a 40% decrease, the transferred value, and a 40% increase.

transferred value by 40% leads to a clearer decrease in SSIM and PSNR for the higher noise levels, whereas increasing it by 40% preserves the performance and in several cases leads to slightly better values. Regarding the LPIPS behaviour, the metric values remain relatively stable when b/a is increased. Decreasing b/a generally has a larger effect and leads to higher LPIPS values at several noise levels, particularly in the higher noise levels, although this behaviour is not observed in all cases. The dependence on λ is also more pronounced and exhibits a metric-dependent trade-off, with SSIM and PSNR generally favoring values around $\lambda/\sigma = 0.4-0.5$, while LPIPS tends to favor slightly stronger regularization. The LPIPS results also show that the parameter values favored by perceptual similarity do not always coincide with those favored by SSIM and PSNR, which further indicates a metric-dependent parameter preference. Overall, the transferred parameter prescription remains within a favorable performance region on Urban100 without dataset-specific retuning.

Finally, we discuss the behavior of the different evaluation metrics in relation to the performance of the proposed model. The proposed method shows a consistent improvement in the classical pixel-wise and structural metrics, while the behavior of LPIPS is less uniform, reflecting the fact that the three metrics capture different aspects of the reconstructed image. Our problem is classical aligned image denoising, where the reconstructed image is expected to remain pixel-wise aligned with the clean reference. For this reason, PSNR and SSIM remain particularly relevant measures in our setting, while LPIPS provides an additional perceptual view based on deep features. In all quantitative evaluations, LPIPS was computed using the standard AlexNet backbone. The L^2 fidelity term of the variational model directly promotes agreement with the observed data and is therefore naturally related to pixel-wise measures such as MSE and PSNR. The regularization term also promotes denoising while preserving important image structures, which can lead to improved structural similarity as measured by SSIM. On the other hand, LPIPS measures similarity in a learned deep-feature space, so improvements in pixel fidelity and structural similarity do not necessarily lead to the same improvement in LPIPS. This interpretation is also consistent with the metric-dependent parameter preference observed in the sensitivity analysis above.

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